



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 03

Experiment Title: Design and Simulation of 8-bit Register using Verilog HDL

Submitted by:

Name: Sakawat Hossain

Student ID: 22201077

Section: AB(AB2)

Submitted to:

Md. Rasheduzzaman

Lecturer

Department of Computer Science and Engineering

Date of Submission:

1. Experiment Name

Design and Simulation of 8-bit Register using Verilog HDL

2. Objective

The objectives of this experiment are:

- To design an 8-bit Register using Verilog HDL.
- To understand the operation of sequential logic circuits.
- To implement data storage functionality using flip-flops.
- To generate RTL schematic using Xilinx ISE.
- To simulate and verify the functionality of the register

3. Apparatus / Hardware & Software Requirements

Software Requirements:

- Xilinx ISE Design Suite 14.7
- ISim Simulator
- Verilog HDL

Hardware Requirements:

- Computer system

4. Circuit Diagram / Schematic

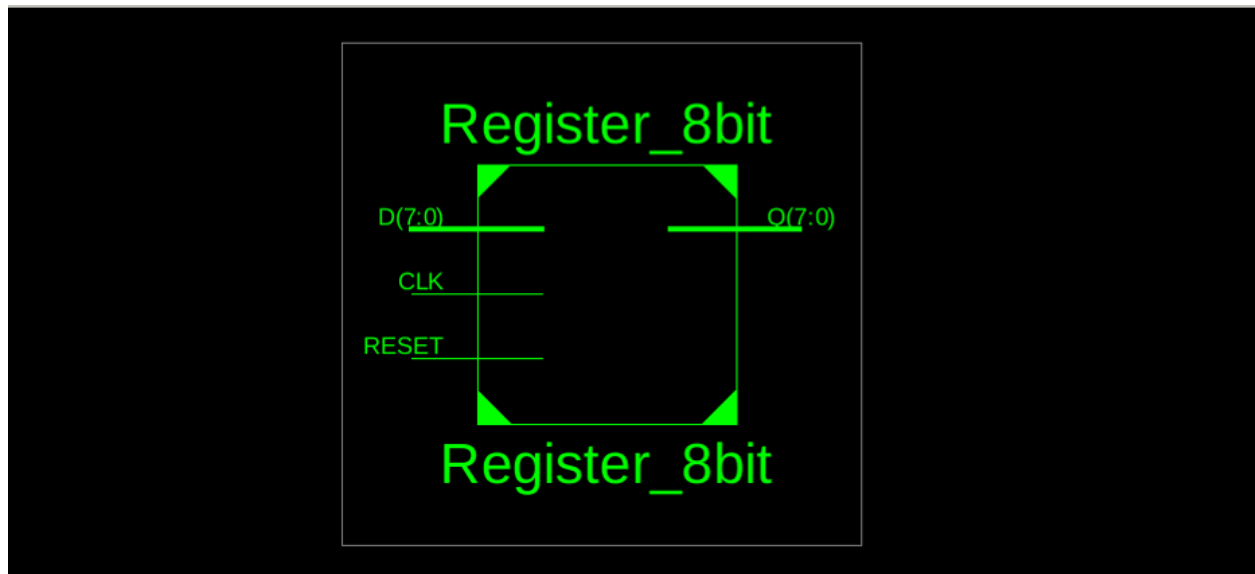


Figure 01: RTL Schematic of 8-bit Register

The RTL schematic represents the functional block of the designed 8-bit register. It contains an 8-bit data input, clock signal, reset input, and 8-bit stored output.

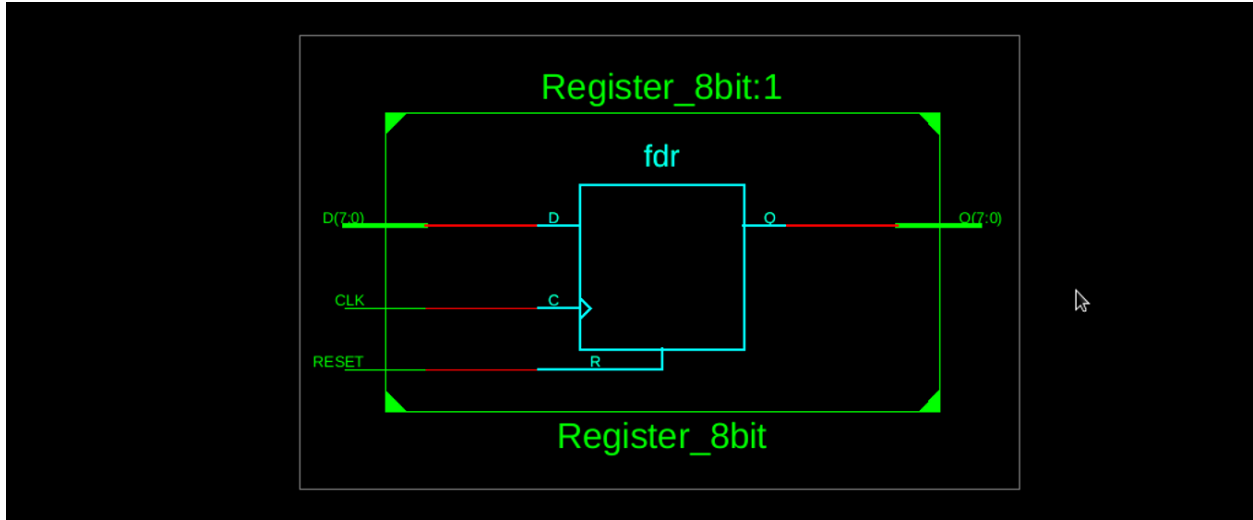


Figure 02: Synthesized Internal Circuit of 8-bit Register

The synthesized circuit shows the internal flip-flop based implementation generated by Xilinx ISE. The 8-bit register is implemented using eight flip-flops.

6. Output Analysis / Observations

The designed 8-bit Register was simulated using ISim simulator.

Different input patterns were applied with clock transitions to verify the storage operation.

Time	RESET	Input D	Output Q
0-20 ns	1	00000000	00000000
20-40 ns	0	10101010	10101010
40-60 ns	0	11110000	11110000
60-80 ns	0	01010101	01010101
80-100 ns	1	11111111	00000000

Table 01: Simulation Test Cases and Output Verification of 8-bit Register

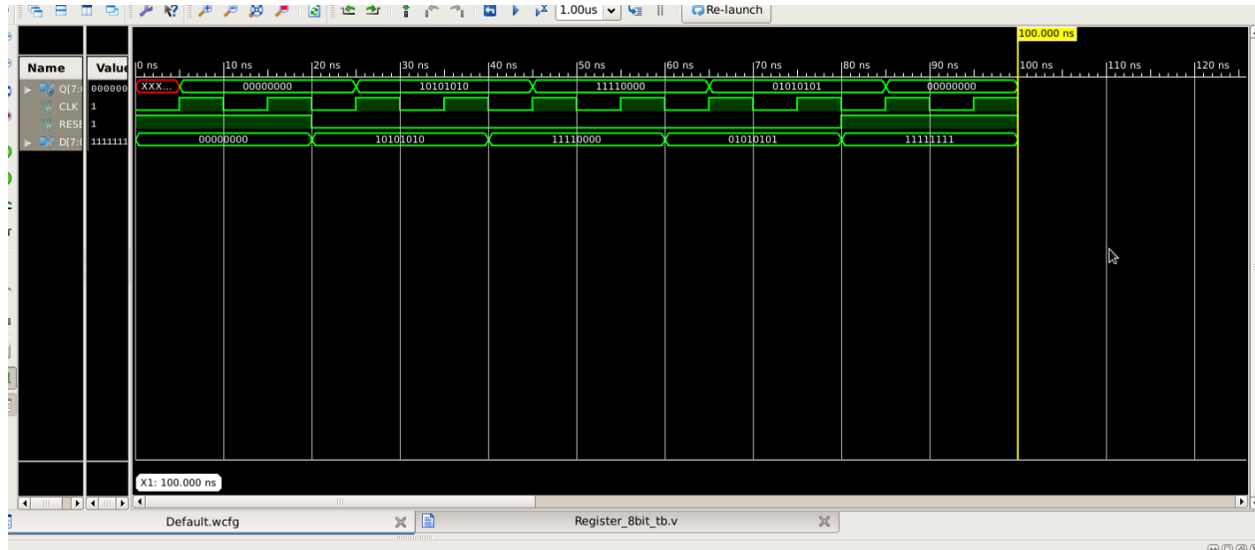


Figure 03: Simulation Waveform of 8-bit Register

The waveform shows that the output follows the input data at the positive edge of the clock and resets to zero when RESET is activated.

7. Result

The 8-bit Register was successfully designed and simulated using Verilog HDL in Xilinx ISE. The simulation results verified that the register correctly stores input data and performs reset operation according to the applied control signals.

The objective of the experiment was successfully achieved.

8. Github Repository Commits

Github Repository Link: https://github.com/Sakhawat025/CSE-450_VLSI_Lab/tree/main/Register_8bit

Github commit Link: https://github.com/Sakhawat025/CSE-450_VLSI_Lab/commits/main/Register_8bit

9. Conclusion

In this experiment, an 8-bit Register was successfully designed using Verilog HDL and simulated using Xilinx ISE. The RTL schematic, synthesized circuit, and simulation waveform were analyzed to verify the correct operation of the sequential circuit. The register successfully stored input data at the clock edge and cleared the output during reset operation.